

Hanzhi "Leo" Liu

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Last updated: May 31, 2026

Education

EMORY UNIVERSITY, Atlanta, GA

Aug 2024 - Current

- **Major:** Physics BS
- **GPA:** 3.97/4.0

Research Experience

EMORY UNIVERSITY, Atlanta, GA

Aug 2025 - Current

Department of Physics
PI: Dr. Justin Burton

- Fall 2025: Using computer vision to extract velocity field from laboratory simulation of ice deformation. Measuring and analyzing the rheological properties of materials serving as glacier analog.

GEORGIA INSTITUTE OF TECHNOLOGY, Atlanta, GA

Summer 2025, Summer 2026

School of Earth and Atmospheric Sciences
PI: Dr. Winnie Chu

- Summer 2025: Constructed an ensemble of thermal models for the Northern Greenland Ice Sheet. Developing and running numerical simulations to investigate the uncertainty in ice sheet temperatures. Financially supported by Emory Pathways Internship Funding.

Conference

AGU25

New Orleans, LA, Dec 2025

- **Liu L.**, Yang, D., Chu, W., MacGregor J. A., "Using Thermal Model Ensemble and Machine Learning to Predict the Basal Temperature of the Northern Greenland Ice Sheet".
- Dawson, E., Christoffersen, M., Yang, D., **Liu, L.**, Chu, W., "Basal Temperatures from Radar Sounding: New Inferences Across East Antarctica".

Awards & Scholarships

Emory Conference Presentation Grants

2025

- Awarded \$750 by Emory's Undergraduate Research Program for traveling to AGU25.

Emory Pathways Internship Funding

2025

- Awarded \$4000 by Emory's Pathways Center, in support of summer undergraduate research assistant role at Georgia Institute of Technology.

National Merit Scholarship

2024

- \$2000 per year scholarship. Sponsored by the National Merit Scholarship Corporation and Emory University. Selected as a National Merit Scholarship Finalist from graduating seniors across the U.S.

Leadership

AGU Cryosphere Canvassing Committee

Member

Feb 2026 - Current

Emory Bike Social

Feb 2026 - Current

Emory University, Atlanta, GA

- Assistant treasurer (Spring 2026).

Outreach & Volunteering

FERNBANK MUSEUM OF NATURAL HISTORY, Atlanta, GA
Volunteer

Jan 2026 - Current

- Assisting museum visitors (mostly elementary school students) with activities at Fernbank's STEAM lab.

CALIFORNIA SCIENCE CENTER, Los Angeles, CA
Guest Services Air and Space Host Volunteer

Oct 2023 - May 2024

- Selected to be a volunteer in charge of educating visitors about the Space Shuttle Endeavour, and other space-related artifacts. Actively engage in educational conversations with visitors about the science and technology behind the Space Shuttle.

Skills

Programming languages: Python (NumPy, matplotlib, pandas, scikit-learn), MATLAB, Julia, Java.

Softwares: Microsoft Office Suite, 3D Modeling (Blender, Substance Painter),

Spoken languages: English (bilingual proficiency), Chinese (native proficiency), Japanese (intermediate working proficiency), French (limited working proficiency).